

Mathematical Modeling and Numerical Simulation Supplement

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1 Q1 (final: v0)

Abstract— A parsimonious ordinary differential equation framework modeling the coupled evolution of neutron star spin period and magnetic inclination angle under vacuum dipole braking, wind enhancement, inclination decay, and accretion-driven spin-up. The model isolates the dynamical signatures of isolated magnetic-dipole evolution versus binary recycling by encoding phase-dependent torque switching directly into the state derivatives, enabling population-level statistical comparison against observed braking indices and characteristic ages.

1.1 Question and Scope

Question. Can the joint distribution of braking index n and characteristic age τ_c statistically distinguish isolated magnetic-dipole evolution from binary recycling populations, given specific torque model constraints?

First executable model: parsimonious ODE integration of single pulsar tracks for isolated magnetic-dipole spin-down with optional inclination decay and wind braking, embedded in a Monte Carlo population synthesis framework. Binary recycling is represented as an accretion-phase switch with an effective spin-up law. The model excludes glitches, magnetospheric state changes, variable moment of inertia, detailed companion ablation, and full binary evolution.

1.2 Model Assumptions and Applicability

- ASM-001: Stellar geometry, electromagnetic constants, and moment of inertia are absorbed into effective braking constants K_{dipole} and K_{acc} . If violated: Derived constants lose direct physical interpretability and may require explicit microphysical coupling.
- ASM-002: Magnetic inclination evolves smoothly via exponential decay without discontinuous glitches or magnetospheric state transitions. If violated: Smooth ODE integration fails to capture abrupt angular momentum transfers, invalidating continuous derivative assumptions.
- ASM-003: Accretion torque is active exclusively during $t < t_{\text{acc}}$ with nonzero $\dot{M}$; isolated dipole torque applies otherwise. If violated: Simultaneous application of opposing torques produces unphysical period trajectories outside the validated regime.

1.3 Symbols and Units

ID	Symbol	Meaning	Unit
S-P	P	Spin period	s
S-alpha	α	Magnetic inclination angle	rad
S-t	t	Integration time	s
S-P0	P_0	Initial spin period	s
S-alpha0	α_0	Initial inclination angle	rad
S-B0	B_0	Surface magnetic field strength	T
S-Kdip	K_{dipole}	Vacuum dipole braking constant	s T^{-2}
S-eta	η	Wind braking efficiency	dimensionless
S-tau	τ_α	Inclination decay timescale	s
S-Mdot	$\dot{M}$	Mass accretion rate	kg s^{-1}
S-Kacc	K_{acc}	Accretion spin-up constant	$\text{s}^2 \text{kg}^{-1}$
S-tacc	t_{acc}	Accretion phase duration	s
S-Tmax	T_{max}	Maximum simulated age	s
S-Pdotmin	$P_{\text{dot,min}}$	Minimum detectable period derivative	dimensionless
S-Nmc	N_{mc}	Monte Carlo sample count	dimensionless

1.4 Mechanistic Equations

$$\frac{dP}{dt} = \begin{cases} -\frac{K_{\text{acc}}\dot{M}}{\max(P, 0.01)} & \text{if } t < t_{\text{acc}} \wedge \dot{M} > 0 \\ +\frac{K_{\text{dipole}}B_0^2 \sin^2(\alpha)(1+\eta)}{\max(P, 0.01)} & \text{otherwise} \end{cases} \quad (1)$$

Q1-EQ-001 where S-P: Spin period; S-t: Integration time; S-Kacc: Accretion spin-up constant; S-Mdot: Mass accretion rate; S-tacc: Accretion phase duration; S-Kdip: Vacuum dipole braking constant; S-B0: Surface magnetic field strength; S-alpha: Magnetic inclination angle; S-eta: Wind braking efficiency.

$$\frac{d\alpha}{dt} = -\frac{\alpha}{\tau_\alpha} \quad (2)$$

Q1-EQ-002 where S-alpha: Magnetic inclination angle; S-t: Integration time; S-tau: Inclination decay timescale.

$$n(t) = \frac{P(t) \cdot \frac{d^2P}{dt^2}}{\left(\frac{dP}{dt}\right)^2} \quad (3)$$

Q1-EQ-003 where S-P: Spin period.

$$\tau_c(t) = \frac{P(t)}{2 \cdot \frac{dP}{dt}} \quad (4)$$

Q1-EQ-004 where S-P: Spin period.

1.5 Initial Conditions, Boundary Conditions, and Constraints

Initial Conditions

- {'state': 'S-P', 'value': 0.5}
- {'state': 'S-alpha', 'value': 0.7853981633974483}

Boundary Conditions

- {'condition': 'Integration terminates when t reaches T_max or |dP/dt| falls below Pdot_min', 'type': 'TERMINATION'}

Objective and Constraints

- {'objective': 'Generate distinct P-Pdot-n trajectories for isolated and recycled populations to enable statistical discrimination.'}
- {'constraint': 'Sign convention: positive dP/dt denotes isolated spin-down; negative dP/dt denotes accretion spin-up. Torque branches must not activate simultaneously.'}

1.6 Algorithm

Algorithm

Input

- Initial conditions (P0, alpha0)
- Approved parameter set
- Scenario configuration

Output

- Time series arrays for P(t), alpha(t)
- Derived diagnostics n(t), tau_c(t)
- Population histograms

Steps

- Initialize state vector with P0 and alpha0
- Select scenario parameter overrides
- Evaluate derivative AST at current time t
- Advance state using fixed-step integrator
- Compute n(t) and tau_c(t) from derivatives
- Check termination criteria against T_max and Pdot_min
- Repeat until termination or N_mc tracks completed

1.7 Parameters, Scenarios, and Numerical Settings

Parameters

- {'symbol_id': 'S-P0', 'value': 0.5}
- {'symbol_id': 'S-alpha0', 'value': 0.7853981633974483}
- {'symbol_id': 'S-B0', 'value': 100000000.0}
- {'symbol_id': 'S-Kdip', 'value': 9.765625e-32}
- {'symbol_id': 'S-eta', 'value': 0.1}
- {'symbol_id': 'S-tau', 'value': 315576000000000.0}
- {'symbol_id': 'S-Mdot', 'value': 630000000000000.0}
- {'symbol_id': 'S-Kacc', 'value': 6.3e-31}
- {'symbol_id': 'S-tacc', 'value': 315576000000000.0}
- {'symbol_id': 'S-Tmax', 'value': 315576000000000.0}
- {'symbol_id': 'S-Pdotmin', 'value': 1e-22}
- {'symbol_id': 'S-Nmc', 'value': 5000.0}

Scenarios

- {'scenario_id': 'isolated_dipole', 'parameter_overrides': {'K_acc': 0.0, 'Mdot': 0.0, 't_acc': 0.0}}
- {'scenario_id': 'binary_recycling', 'parameter_overrides': {'K_acc': 5e-31}}

Solver family. Explicit Runge-Kutta 4th Order.

1.8 Parameter Provenance and Applicability

Approved parameter-set identity. 0f0174315925c8e5b143507cd77c961e1becdad36afebaa31e7f6df12d27946c.

- initial_period (P0) = 0.5 s; role BOUNDARY_CONDITION; status APPROVED_USER_INPUT; source: Birth and Evolution of Isolated Radio Pulsars; DOI 10.1086/501516; document Q1-FGK2006; locator: preprint; applicability: required_condition_1=For isolated or recycled neutron stars at the beginning of the simulated spin-down phase, with period measured in seconds and no glitch-induced period jump at the initial epoch.. uncertainty: not reported.
- initial_inclination (alpha0) = 0.7853981633974483 rad; role BOUNDARY_CONDITION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=For dipole torque models with approximately aligned magnetic and rotational axes represented by an angle in radians; treated as the starting value before inclination decay.. uncertainty: not reported.

- `initial_surface_magnetic_field (B0)` = 100000000.0 T; role `BOUNDARY_CONDITION`; status `APPROVED_USER_INPUT`; source: Binary and Millisecond Pulsars; DOI 10.12942/lrr-2008-8; document Q1-LORIMER2008; locator: `review_article`; applicability: `required_condition_1=For isolated neutron stars with approximately dipolar surface fields inferred from timing data, expressed in tesla; not intended for magnetar-like or strongly multipolar fields..` uncertainty: not reported.
- `vacuum_dipole_braking_constant (K_dipole)` = $9.765625 \times 10^{-32} \text{ s } T^{-2}$; role `MODEL_ASSUMPTION`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1=For the chosen effective spin-down law with magnetic field in tesla and period in seconds, assuming fixed stellar geometry and no unresolved internal braking physics..` uncertainty: not reported.
- `wind_braking_efficiency (eta)` = 0.1 dimensionless; role `MODEL_ASSUMPTION`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1=For pulsars with magnetospheric wind torque included as a multiplicative enhancement of the dipole term; scenario switches may disable the wind contribution..` uncertainty: not reported.
- `inclination_decay_timescale (tau_alpha)` = 31557600000000.0 s; role `MODEL_ASSUMPTION`; status `APPROVED_USER_INPUT`; source: Binary and Millisecond Pulsars; DOI 10.12942/lrr-2008-8; document Q1-LORIMER2008; locator: `review_article`; applicability: `required_condition_1=For isolated pulsars modeled with smooth inclination decay, expressed in seconds, excluding glitches and abrupt magnetospheric state changes..` uncertainty: not reported.
- `accretion_rate (Mdot)` = 63000000000000.0 $\text{kg } s^{-1}$; role `SCENARIO_INPUT`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1=For recycled pulsars in a steady disk-accretion phase, expressed in kilograms per second, ignoring transient outbursts and companion ablation..` uncertainty: not reported.
- `accretion_braking_constant (K_acc)` = $6.3 \times 10^{-31} \text{ s }^2 \text{ kg}^{-1}$; role `SCENARIO_INPUT`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1=For the effective accretion law using accretion rate in kilograms per second and dimensionless period derivative; active only during the accretion phase..` uncertainty: not reported.
- `accretion_duration (t_acc)` = 31557600000000.0 s; role `SCENARIO_INPUT`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1=For binary recycling tracks represented by a finite accretion interval followed by isolated spin-down, expressed in seconds..` uncertainty: not reported.
- `maximum_simulated_age (T_max)` = 31557600000000.0 s; role `BOUNDARY_CONDITION`; status `APPROVED_USER_INPUT`; source: Birth and Evolution of Isolated Radio Pulsars; DOI 10.1086/501516; document Q1-FGK2006; locator: `preprint`; applicability: `required_condition_1=Number of tracks to simulate; not a physical observable unless tied to survey selection or evolutionary assumptions..` uncertainty: not reported.
- `minimum_detectable_period_derivative (Pdot_min)` = 1×10^{-22} dimensionless; role `BOUNDARY_CONDITION`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL AS-`

SUMPTION (not a measured or literature value); applicability: required_condition_1=Survey-dependent detection boundary for timing observations; used as a stopping condition rather than a physical spin-down parameter.. uncertainty: not reported.

- monte_carlo_sample_count (N_mc) = 5000.0 dimensionless; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=Numerical control for population synthesis; chosen for computational budget and statistical resolution rather than physical measurement.. uncertainty: not reported.

1.9 Numerical Validation and Model-Internal Results

Validation Plan

- {'method': 'Analytical limit check: verify $d\alpha/dt$ converges to zero as t approaches infinity', 'metric': 'Relative error $< 1e-6$ '}
- {'method': 'Scenario isolation test: confirm isolated_dipole yields strictly positive dP/dt throughout', 'metric': 'Sign consistency across all t '}
- {'method': 'Recycling transition test: verify dP/dt switches sign at t_{acc} and returns to positive post-accretion', 'metric': 'Branch activation count matches scenario definition'}
- Execution sim-ef92f26db7c14196aaab75759811335e; NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL; relation CONSTRAINED; summary: Within the audited reduced ODE, the isolated track increases from $P=0.500$ s to 0.662 s, whereas the recycling track reaches $P_{min}=0.226$ s during accretion and ends at 0.281 s after resumed dipole spin-down. This establishes model-internal directional separation, but does not by itself validate population-level $n\text{-}\tau_c$ discrimination because the executable MathIR contains two deterministic tracks rather than the documented Monte Carlo outer loop.

1.10 Hypothesis Iteration Lineage

- v0: CONSTRAINED (Within the audited reduced ODE, the isolated track increases from $P=0.500$ s to 0.662 s, whereas the recycling track reaches $P_{min}=0.226$ s during accretion and ends at 0.281 s after resumed dipole spin-down. This establishes model-internal directional separation, but does not by itself validate population-level $n\text{-}\tau_c$ discrimination because the executable MathIR contains two deterministic tracks rather than the documented Monte Carlo outer loop.)

1.11 Limitations, Non-Empirical Disclosure, and References

All values in this section are model-internal numerical simulations (NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL), not empirical observations.

Limitations

- Excludes glitch events, magnetospheric state transitions, and variable moment of inertia.
- Assumes fixed stellar geometry and absorbs unresolved electromagnetic factors into effective constants.

- Observational selection effects and survey detection thresholds are not dynamically coupled to the simulation loop.

References

- {'citation': 'Feng, G., et al. (2006). Birth and Evolution of Isolated Radio Pulsars. *ApJ*, 646, 381. DOI: 10.1086/501516', 'source': 'Q1-FGK2006'}
- {'citation': 'Lorimer, D. R. (2008). Binary and Millisecond Pulsars. *Living Rev. Relativity*, 11, 8. DOI: 10.12942/lrr-2008-8', 'source': 'Q1-LORIMER2008'}

2 Q2 (final: v0)

Abstract— A reduced-order ordinary differential equation initial-value model simulates the thermal evolution of isolated massive neutron stars under magnetic-dipole spin-down. The system tracks core temperature, surface temperature, central density, spin frequency, and a phase-fraction proxy. Scenario-specific switches activate latent-heat release and superfluid-modified cooling proxies when spin-down compression crosses a critical density threshold, enabling direct comparison of standard cooling, phase-transition cooling, and superfluid-enhanced phase-transition tracks.

2.1 Question and Scope

Question. How does the surface temperature evolution differ between massive neutron stars undergoing spin-down induced core compression and those without, and can this distinguish the phase-transition mechanism from standard cooling?

Isolated massive neutron star thermal evolution under magnetic-dipole spin-down, with a reduced initial-value model for core temperature, surface temperature, spin frequency, and central density. The model excludes binary recycling, companion ablation, detailed crustal heat blanketing, resolved neutrino microphysics, full equation-of-state integration, and non-spherical structure. Phase transition and superfluidity are represented through explicit scenario selection and effective coefficients.

2.2 Model Assumptions and Applicability

- ASM-001: Neutron stars are treated as spherically symmetric, isolated systems with no binary accretion torque or magnetic field evolution. If violated: Spin-down torque and rotational compression proxies become inaccurate; observed trajectories would diverge from modeled tracks.
- ASM-002: Detailed neutrino emissivity processes and heat-capacity microphysics are absorbed into a single effective cooling timescale and scenario-dependent coefficients. If violated: Temperature relaxation rates would misrepresent actual stellar cooling physics, invalidating comparative scenario analysis.
- ASM-003: Moment of inertia variations are fully parameterized into the rotational compression coefficient rather than solved dynamically. If violated: Central density evolution would decouple from realistic spin-down-induced structural changes.

2.3 Symbols and Units

ID	Symbol	Meaning	Unit
SYM-001	$\Omega(t)$	Angular spin frequency	rad s^{-1}
SYM-002	$\rho_c(t)$	Central density	kg m^{-3}
SYM-003	$T_{\text{core}}(t)$	Core temperature	K
SYM-004	$x(t)$	Phase fraction proxy	dimensionless
SYM-005	$T_{\text{surface}}(t)$	Surface temperature	K
SYM-006	M_{NS}	Gravitational mass of the neutron star	solar mass
SYM-007	T_{c0}	Initial core temperature	K
SYM-008	Ω_0	Initial angular spin frequency	rad s^{-1}
SYM-009	ρ_{c0}	Initial central density	kg m^{-3}
SYM-010	$\dot{P}_0$	Initial spin period derivative	s s^{-1}
SYM-011	ρ_{crit}	Critical density for phase transition	kg m^{-3}
SYM-012	L_{latent}	Effective latent-heat temperature proxy	K
SYM-013	S	Scenario selector switch	dimensionless
SYM-014	n	Braking index	dimensionless
SYM-015	τ_{cool}	Effective cooling timescale	s
SYM-016	α_{cs}	Core-to-surface temperature conversion coefficient	dimensionless
SYM-017	κ_{rot}	Rotational compression coefficient	$\text{kg s}^2 \text{m}^{-3}$

2.4 Mechanistic Equations

$$d\Omega/dt = -(\dot{P}_0/2\pi) \cdot \Omega_0^{2-n} \cdot \Omega^n \quad (5)$$

Q2-EQ-001 where SYM-001: Angular spin frequency; SYM-008: Initial angular spin frequency; SYM-010: Initial spin period derivative; SYM-014: Braking index.

$$d\rho_c/dt = -\kappa_{\text{rot}} \cdot \Omega \cdot (d\Omega/dt) \cdot (M_{\text{NS}}/1.42) \quad (6)$$

Q2-EQ-002 where SYM-002: Central density; SYM-001: Angular spin frequency; SYM-017: Rotational compression coefficient; SYM-006: Gravitational mass of the neutron star.

$$dx/dt = \begin{cases} (1-x)/(0.01\tau_{\text{cool}}) & S \geq 1 \wedge \rho_c \geq \rho_{\text{crit}} \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

Q2-EQ-003 where SYM-004: Phase fraction proxy; SYM-013: Scenario selector switch; SYM-002: Central density; SYM-011: Critical density for phase transition; SYM-015: Effective cooling timescale.

$$dT_{\text{core}}/dt = -\gamma \cdot T_{\text{core}}/\tau_{\text{cool}} + L_{\text{latent}} \cdot dx/dt, \gamma = \begin{cases} 1 & S < 2 \\ 0.5 & S \geq 2 \end{cases} \quad (8)$$

Q2-EQ-004 where SYM-003: Core temperature; SYM-013: Scenario selector switch; SYM-015: Effective cooling timescale; SYM-012: Effective latent-heat temperature proxy; SYM-004: Phase fraction proxy.

$$dT_{\text{surface}}/dt = \alpha_{cs} \cdot dT_{\text{core}}/dt \quad (9)$$

Q2-EQ-005 where SYM-005: Surface temperature; SYM-016: Core-to-surface temperature conversion coefficient; SYM-003: Core temperature.

2.5 Initial Conditions, Boundary Conditions, and Constraints

Initial Conditions

- {'condition_id': 'IC-001', 'state': 'Omega', 'value': 30.44178928}
- {'condition_id': 'IC-002', 'state': 'rho_c', 'value': 4.5e+17}
- {'condition_id': 'IC-003', 'state': 'T_core', 'value': 1000000000.0}
- {'condition_id': 'IC-004', 'state': 'phase_fraction', 'value': 0.0}
- {'condition_id': 'IC-005', 'state': 'T_surface', 'value': 1000000.0}

Boundary Conditions

- {'condition_id': 'BC-001', 'description': 'Surface temperature initialized as linear projection of core temperature via alpha_cs.', 'symbol_refs': ['SYM-005', 'SYM-003', 'SYM-016']}
- {'condition_id': 'BC-002', 'description': 'Central density seeded below transition threshold to permit spin-down crossing.', 'symbol_refs': ['SYM-002', 'SYM-011']}

Objective and Constraints

- {'constraint_id': 'OBJ-001', 'description': 'Maximize physical fidelity within reduced-order framework while maintaining computational tractability for scenario comparison.'}
- {'constraint_id': 'OBJ-002', 'description': 'Ensure all temperatures, densities, and phase fractions remain non-negative throughout integration.'}
- {'constraint_id': 'OBJ-003', 'description': 'Limit output trajectory points to 2000 or fewer by enforcing appropriate maximum step size relative to total simulation horizon.'}

2.6 Algorithm

Algorithm

Input

- Parameter set values
- Scenario selector override
- Time span bounds

Output

- State trajectories over time
- Surface temperature vs age tracks

Steps

- Initialize state vector from approved initial conditions and parameter values.
- Evaluate scenario selector to determine active cooling proxy and latent-heat activation logic.
- Integrate coupled ODE system using stiff adaptive solver over defined time span.
- Apply conditional branching for phase fraction evolution based on central density threshold.
- Compute surface temperature trajectory from core temperature derivative and conversion coefficient.
- Export synchronized time-series arrays for post-processing and observational comparison.

2.7 Parameters, Scenarios, and Numerical Settings

Parameters

- {'parameter_id': 'neutron_star_mass', 'mathir_symbol': 'M_NS', 'value': 1.42, 'unit': 'solar_mass', 'role': 'SCENARIO_INPUT'}
- {'parameter_id': 'initial_core_temperature', 'mathir_symbol': 'T_c0', 'value': 1000000000.0, 'unit': 'K', 'role': 'BOUNDARY_CONDITION'}
- {'parameter_id': 'initial_spin_frequency', 'mathir_symbol': 'Omega0', 'value': 30.44178928, 'unit': 'rad_s^-1', 'role': 'SCENARIO_INPUT'}
- {'parameter_id': 'initial_central_density', 'mathir_symbol': 'rho_c0', 'value': 4.5e+17, 'unit': 'kg_m^-3', 'role': 'BOUNDARY_CONDITION'}
- {'parameter_id': 'initial_spin_period_derivative', 'mathir_symbol': 'Pdot0', 'value': 9.7228e-13, 'unit': 's_s^-1', 'role': 'SCENARIO_INPUT'}
- {'parameter_id': 'phase_transition_critical_density', 'mathir_symbol': 'rho_crit', 'value': 5.12e+17, 'unit': 'kg_m^-3', 'role': 'MATERIAL_PROPERTY'}
- {'parameter_id': 'effective_latent_heat_temperature_release', 'mathir_symbol': 'L_latent', 'value': 50000000.0, 'unit': 'K', 'role': 'MATERIAL_PROPERTY'}
- {'parameter_id': 'scenario_selector', 'mathir_symbol': 'scenario', 'value': 1.0, 'unit': 'dimensionless', 'role': 'SCENARIO_INPUT'}
- {'parameter_id': 'braking_index', 'mathir_symbol': 'n', 'value': 3.15, 'unit': 'dimensionless', 'role': 'MODEL_ASSUMPTION'}
- {'parameter_id': 'effective_cooling_timescale', 'mathir_symbol': 'tau_cool', 'value': 3155760000000.0, 'unit': 's', 'role': 'MODEL_ASSUMPTION'}
- {'parameter_id': 'core_surface_conversion_coefficient', 'mathir_symbol': 'alpha_cs', 'value': 0.001, 'unit': 'dimensionless', 'role': 'BOUNDARY_CONDITION'}
- {'parameter_id': 'rotational_compression_coefficient', 'mathir_symbol': 'kappa_rot', 'value': 15000000000000.0, 'unit': 'kg_s^2_m^-3', 'role': 'MODEL_ASSUMPTION'}

Scenarios

- {'scenario_id': 'standard_cooling', 'parameter_overrides': {'scenario': 0.0}, 'description': 'Standard cooling without phase transition or superfluid modification.'}
- {'scenario_id': 'phase_transition', 'parameter_overrides': {'scenario': 1.0}, 'description': 'Phase transition cooling with latent heat release upon density threshold crossing.'}
- {'scenario_id': 'phase_transition_superfluid', 'parameter_overrides': {'scenario': 2.0}, 'description': 'Phase transition cooling with modified neutrino emissivity proxy (reduced cooling factor).')}

Solver family. Stiff ODE IVP integrator (e.g., LSODA or Radau).

2.8 Parameter Provenance and Applicability

Approved parameter-set identity. 8960cd825beacfd9059c043b2efe1da4859a4586f8ae0739d68d043e5e953cc1.

- `neutron_star_mass (M_NS)` = 1.42 solar_mass; role SCENARIO_INPUT; status APPROVED_USER_INPUT; source: Signature of Deconfinement with Spin-down Compression in Cooling Hybrid Stars; DOI 10.1088/0004-637X/694/2/1019; document Q2-STEJNER2009; locator: local_document; applicability: required_condition_1=Isolated massive neutron star; spherical symmetry; non-recycled spin-down history; no binary accretion torque.. uncertainty: not reported.
- `initial_core_temperature (T_c0)` = 1000000000.0 K; role BOUNDARY_CONDITION; status APPROVED_USER_INPUT; source: Signature of Deconfinement with Spin-down Compression in Cooling Hybrid Stars; DOI 10.1088/0004-637X/694/2/1019; document Q2-STEJNER2009; locator: local_document; applicability: required_condition_1=Young isolated neutron star at the beginning of the cooling track; model assumes initial thermal state before long-term cooling; no crustal decoupling resolved.. uncertainty: not reported.
- `initial_spin_frequency (Omega0)` = 30.44178928 rad_s⁻¹; role SCENARIO_INPUT; status APPROVED_USER_INPUT; source: Controlled excerpt from On the magnetic field evolution of PSR J1640-4631; DOI 10.3847/1538-4357/aa68e6; document Q2-GAO2017-PDOT-EXCERPT; locator: user_provided_local_document; applicability: required_condition_1=Birth spin state of an isolated neutron star; magnetic dipole spin-down; no binary recycling or accretion-driven spin-up.. uncertainty: not reported.
- `initial_central_density (rho_c0)` = 4.5e+17 kg_m⁻³; role BOUNDARY_CONDITION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=Adopted for the first executable parsimonious model; must be internally consistent with the selected mass and EOS proxy; no full EOS integration is resolved.. uncertainty: not reported.
- `initial_spin_period_derivative (Pdot0)` = 9.7228e-13 s_s⁻¹; role SCENARIO_INPUT; status APPROVED_USER_INPUT; source: Controlled excerpt from On the magnetic field evolution of PSR J1640-4631; DOI 10.3847/1538-4357/aa68e6; document Q2-GAO2017-PDOT-EXCERPT; locator: user_provided_local_document; applicability: required_condition_1=Isolated pulsar timing measurement; magnetic dipole braking; no accretion torque; no significant magnetic field evolution resolved.. uncertainty: not reported.

- `phase_transition_critical_density` (ρ_{crit}) = $5.12 \times 10^{17} \text{ kg m}^{-3}$; role MATERIAL_PROPERTY; status APPROVED_USER_INPUT; source: Signature of Deconfinement with Spin-down Compression in Cooling Hybrid Stars; DOI 10.1088/0004-637X/694/2/1019; document Q2-STEJNER2009; locator: local_document; applicability: required_condition_1=Beta-equilibrated dense matter composition; EOS-dependent transition; relevant temperature and pressure regime for neutron star interiors.. uncertainty: not reported.
- `effective_latent_heat_temperature_release` (L_{latent}) = 50000000.0 K; role MATERIAL_PROPERTY; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=EOS-dependent phase transition; beta-equilibrated dense matter; conversion from microscopic latent heat to a temperature proxy is a model assumption; no resolved heat-capacity microphysics.. uncertainty: not reported.
- `scenario_selector` (scenario) = 1.0 dimensionless; role SCENARIO_INPUT; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=Executable parsimonious model; scenario choice is explicit and not represented as a literature value.. uncertainty: not reported.
- `braking_index` (n) = 3.15 dimensionless; role MODEL_ASSUMPTION; status APPROVED_USER_INPUT; source: Controlled excerpt from On the magnetic field evolution of PSR J1640-4631; DOI 10.3847/1538-4357/aa68e6; document Q2-GAO2017-PDOT-EXCERPT; locator: user_provided_local_document; applicability: required_condition_1=Adopted spin-down law assumption; no resolved magnetic field evolution; no accretion torque.. uncertainty: not reported.
- `effective_cooling_timescale` (τ_{cool}) = 3155760000000.0 s; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=Parsimonious first executable model; detailed Urca processes, modified Urca processes, direct Urca thresholds, and superfluid pairing are not resolved separately.. uncertainty: not reported.
- `core_surface_conversion_coefficient` (α_{cs}) = 0.001 dimensionless; role BOUNDARY_CONDITION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=Spherical symmetry; simplified atmosphere boundary; no resolved radiative transfer or crustal thermal blanketing.. uncertainty: not reported.
- `rotational_compression_coefficient` (κ_{rot}) = $15000000000000.0 \text{ kg s}^{-2} \text{ m}^{-3}$; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=Small spin-down induced density changes; no full rotating EOS integration; moment of inertia response is absorbed into the proxy.. uncertainty: not reported.

2.9 Numerical Validation and Model-Internal Results

Validation Plan

- {'plan_id': 'VAL-001', 'method': 'Cross-scenario consistency check: verify that changing scenario selector alters at least one derivative term or state initialization.'}
- {'plan_id': 'VAL-002', 'method': 'Boundary limit test: confirm surface temperature remains positive and bounded for all three approved scenarios.'}

- {'plan_id': 'VAL-003', 'method': 'Observational bracketing: compare simulated $T_s(t)$ tracks against published X-ray cooling curves for isolated massive pulsars.'}
- Execution sim-05200df7f99b4a56b313fb266667735f; NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL; relation SUPPORTED_WITHIN_MODEL; summary: The audited reduced ODE crosses $\rho_{\text{crit}}=5.12\text{e}17 \text{ kg/m}^3$ at approximately $9.85\text{e}11 \text{ s}$ in all spin-down cases. At $3\tau_{\text{cool}}$, the standard, phase-transition, and phase-transition-plus-superfluid scenarios yield surface temperatures of about $4.98\text{e}4 \text{ K}$, $5.32\text{e}4 \text{ K}$, and $2.36\text{e}5 \text{ K}$, respectively; phase_fraction reaches 1 only in the two transition scenarios. The mechanism therefore produces distinguishable model-internal thermal tracks, conditional on the effective cooling, latent-heat, and compression proxies.

2.10 Hypothesis Iteration Lineage

- v0: SUPPORTED_WITHIN_MODEL (The audited reduced ODE crosses $\rho_{\text{crit}}=5.12\text{e}17 \text{ kg/m}^3$ at approximately $9.85\text{e}11 \text{ s}$ in all spin-down cases. At $3\tau_{\text{cool}}$, the standard, phase-transition, and phase-transition-plus-superfluid scenarios yield surface temperatures of about $4.98\text{e}4 \text{ K}$, $5.32\text{e}4 \text{ K}$, and $2.36\text{e}5 \text{ K}$, respectively; phase_fraction reaches 1 only in the two transition scenarios. The mechanism therefore produces distinguishable model-internal thermal tracks, conditional on the effective cooling, latent-heat, and compression proxies.)

2.11 Limitations, Non-Empirical Disclosure, and References

All values in this section are model-internal numerical simulations (NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL), not empirical observations.

Limitations

- Highly sensitive to the choice of EOS and phase transition parameters; observational uncertainties in distance and radius affect temperature inference.
- Assumes spherical symmetry and ignores binary recycling, companion-ablation evolution, and detailed crustal heat blanketing.
- No resolved neutrino microphysics or full equation-of-state integration; relies on effective proxy coefficients for first executable run.

References

- {'ref_id': 'REF-001', 'citation': 'Stejner et al. (2009), Signature of Deconfinement with Spin-down Compression in Cooling Hybrid Stars, ApJ, 694, 1019.'}
- {'ref_id': 'REF-002', 'citation': 'Gao et al. (2017), On the magnetic field evolution of PSR J1640-4631, ApJ, 840, 12.'}